Project Report

Credit Card Approval Prediction System

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CreditGuard AI: Credit Card Approval Prediction System

Final Project Report

Abstract

This project presents CreditGuard AI, an end-to-end supervised machine learning platform designed to automate credit card risk assessment and automate card approvals. Using a dataset consisting of applicant demographics and monthly payment records, the system maps default delinquency risk (>= 60 days delinquent status) and deploys a lightweight, production-grade Flask REST server. The system prioritized Recall to safeguard financial resources from credit default and successfully generalized under severe class imbalance.

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1. Introduction

Modern banking requires quick, automated credit decisions. Traditional underwriting relies on slow, manual reviews. Implementing Supervised Binary Classifiers enables institutions to automate risk ratings.

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2. Problem Statement

Credit card approval skew features a high volume of low-risk approvals (Class 0: 92.5%) and a minor, highly-consequential cluster of high-risk delinquents (Class 1: 7.5%). Standard ML estimators overfit the majority class, risking severe financial losses from undetected default cases.

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3. System Requirements

- Data Partitioning: Stratified 80/20 train/test split.
- Imbalance Strategy: Balanced Random Oversampling inside training splits.
- Python Runtime: Python 3.10 / 3.13.
- Deployment Platform: Vercel and Docker container engines.

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4. Exploratory Data Analysis & Cleaning

- Income Skew: Heavy log-normal tailing corrected using 1.5 IQR boundary limits capping.
- Logical Ranges: Cleansed negative incomes and child counts, and set family sizes to >= 1.

- Null Fields: Mode-imputed missing values inside `OCCUPATION_TYPE` to avoid list omissions.

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5. Feature Engineering & Selection

- Discretionary Income: Formulated `INCOME_PER_MEMBER`.
- Financial Stability Score: Scaled asset index (0 to 3) representing car/property ownership and high income.
- Selection Gini Rankings: 42 features selected based on Random Forest Importance and Mutual Information.

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6. Model Training & Comparison

We evaluated Logistic Regression, Decision Tree, Random Forest, and XGBoost models:

- Logistic Regression: F1-Score of 0.2387, ROC-AUC of 0.7409, Recall of 66.67%.
- XGBoost: F1-Score of 0.2047, ROC-AUC of 0.7001, Recall of 17.33%.
- Random Forest: F1-Score of 0.1930, ROC-AUC of 0.7174, Recall of 14.67%.

Logistic Regression was selected as the Deployed Model due to its balanced recall, preventing default risk under-predictions.

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7. Production Web Application & Deployment

- Flask Server: Uses Blueprints, WTForms backend validation, and custom 404/500 handlers.
- REST Endpoints: `/health` (system diagnostics) and `POST /api/predict` (scoring API).
- Containerization: Configured `Dockerfile` and `docker-compose.yml`.

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8. References

1. Kaggle Credit Card Approval Dataset: [Link](https://www.kaggle.com/datasets/rikdifos/credit-card-approval-prediction).

2. Scikit-Learn Documentation: [Link](https://scikit-learn.org).